What does the band above do differently?

The leak ledger's causes on 74,000 Lichess 10-minute games, his level against 200 points above

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Abstract

Players 200 points above his level give away 5.55 fewer points a game than players at his level (4.41 to 6.70), on 74,000 Lichess 10-minute games built with the app's own code. Six of the leak ledger's ten causes separate the two bands. Ranked by the difference, they are tense middlegames, the move after the opponent's serious error, tense openings, tense positions answered in under five seconds, tense endgames and quiet openings. Tense positions carry more than the whole difference (119%). The band above gives away more a game in quiet middlegames and endgames only because its games are longer: per move it gives away less or the same on every cause. Serious errors carry 94% of the difference, and hanging a piece 43%; his level hangs one once every 1.2 games, the band above once every 1.4. The rule's fit of the result on the points given away finds the app's 0.6 discount too high on every cause. But that fit leaves out the opponent's points, which rise and fall with the player's. With them in, in a check added after the results were read, a point given away costs about a point of result (1.05). The data give no reason to lower the 0.6. In his own games, tense endgames take a larger share of his points than at either level, and he allows a mate about 1.6 times as often as players at his level.

1 The question

The app's ONE focus is a leak from the metrics service's ledger (`leakLedgerOf`, at commit `d8b771238c`): the points a game the player gives away on one kind of move, against what his opponents give away on the same kind of move in the same games. The kinds of move are the ledger's ten causes: the phase and whether the position was tense, under a minute on the clock, a tense position answered in under five seconds, the move after the opponent's serious error, and moves from move 40. A point is a percent of a game, the drop in expected score across the move, and a leak is priced in Elo as 7 times a hand-set discount of 0.6 times the gap: the share of a point given away that a fix would bank. Two things the ledger does not measure: where players a band above his give away less than players at his level, and how much of a point given away shows in the result. So:

1. On which causes do players 200 points above his level (1450–1649 on Lichess) give away fewer points a game than players at his level (1250–1449), and which carry the difference?
2. How much does a point given away on each cause cost in the result, against the app's 0.6?
3. How often do players at each level hang a piece or allow a mate, and what does it cost them?

2 The data

The games are the 74,000 rated 10-minute Lichess games the think-time study kept from 2026-09 (CC0), with Lichess's evaluation on every move and the clock. Each was built once with the app's own code through the lab's bridge: the metrics service's `primaryFacts` with the routine's ports, and the review service's reader for its motif tags. Each game gives two player-games, one for each player as "me"; Black's facts are derived from White's build with the evaluations and the result turned around, and on 800 games they matched a real Black build, move facts, score and tags. A player-game's band is the player's Lichess rating (Table 1). His chess.com rapid rating comes to the middle of his band by the app's default conversion.

player's rating	player-games	games	slope k ($\times 10^{-3}$)	plies
850–1049	13,282	8,310	1.76	60
1050–1249	21,692	13,937	2.01	62
1250–1449 (his)	39,903	24,168	2.22	64
1450–1649 (above)	34,973	20,589	2.45	66
1650–1849	15,268	9,353	2.72	69
1850–2049	12,424	8,025	3.04	73

Table 1: The bands. The slope is the band curve's, fitted on the band's own player-games; plies are a game's mean length.

3 The method

Every measure is the app's. The points a player gives away on a cause are the method service's `causeLoss` for that cause: the drop in expected score across each of his moves of that kind, on the band curve, floored at zero. One yardstick prices every band: the curve's slope fitted by the metrics service's `fitWinCurve` on his band's player-games ($k = 2.22 \times 10^{-3}$ per centipawn, from 2,276,949 positions; the Lichess curve's is 3.68), so the same move costs the same points at every level. The ledger's first seven causes, the six phase-by-position ones and the low clock, split every move between them, so their differences add up to the difference on every move.

Error kinds come from the review reader's motif tags on the moves it calls a mistake or a blunder, read from the evaluations alone: with no engine line, a move that leaves one of the mover's pieces newly hanging is *hanging a piece*, and the evaluation or the board gives *allowing mate in one* and *allowing a forced mate* (in two to five). The three kinds and the untagged rest split the serious errors.

Means are over player-games. Intervals come from 2,000 bootstrap draws over games (a game's two player-games share its Poisson weight; seed 20261014), the 95% percentile interval, with a p-value of twice the smaller share of draws on either side of zero.

The rule, committed before any result was computed. (1) A cause *separates the bands* when the band above's mean points a game on it minus his band's, Δ_c, is below zero with a Holm-adjusted p under 0.05 over the ten causes; the causes that separate them, ranked by Δ_c, are the priorities the data give his band. (2) Over his band's player-games, a least squares fit of the result (0, 50 or 100) on the points given away on a cause and the rating difference gives δ_c, minus the slope on the points: the share of a point given away that shows in the result. The app's 0.6 is *too high* for a cause when δ_c's interval lies below it, *too low* when above, and *within* otherwise.

Four checks were added after the results were read, and each is marked where it appears: points per move, the error kinds' share of the difference, the discount fitted with the opponent's points, and his own games' split. They test nothing and carry no rule; they say what the registered numbers mean.

4 Results

4.1 What separates the bands

Over every move, the band above gives away 103.1 points a game and his band 108.6, a difference of −5.55 (−6.70 to −4.41; Table 2). Six of the ten causes separate the bands. Ranked by the difference, they are tense middlegames, the move after the opponent's serious error, tense openings, tense positions answered in under five seconds, tense endgames and quiet openings. The first two cost about three points a game each: tense middlegames −3.05 and the move after the opponent's serious error −2.97.

The seven causes that split every move add up to the difference on every move, and the tense ones carry more than all of it: 45% in the opening, 55% in the middlegame and 19% in the endgame, 119% together. Quiet openings add 13%. The band above gives away *more* a game in quiet middlegames (+0.97), quiet endgames (+0.56), under a minute on the clock (+0.24) and from move 40 (+0.85), the four causes that do not separate the bands; the first three give back 32% of the difference.

	points a game			Holm		share of
cause	his band	above	Δ	p	separates	the difference
opening, tense	13.8	11.3	−2.51 (−2.79, −2.24)	0.010	yes	45% (37, 58)
opening, quiet	12.1	11.3	−0.72 (−0.89, −0.53)	0.010	yes	13% (10, 18)
middlegame, tense	38.2	35.1	−3.05 (−3.67, −2.42)	0.010	yes	55% (45, 67)
middlegame, quiet	14.8	15.7	+0.97 (+0.66, +1.26)	0.010	no	−18% (−26, −11)
endgame, tense	18.9	17.9	−1.05 (−1.60, −0.48)	0.010	yes	19% (10, 26)
endgame, quiet	8.8	9.4	+0.56 (+0.18, +0.93)	0.010	no	−10% (−20, −3)
under a minute left	2.2	2.4	+0.24 (+0.02, +0.46)	0.033	no	−4% (−9, −0)
tense, under 5 seconds	23.8	22.4	−1.40 (−1.83, −0.97)	0.010	yes	
after the opponent's error	28.5	25.5	−2.97 (−3.56, −2.35)	0.010	yes	
from move 40	8.9	9.8	+0.85 (+0.31, +1.39)	0.010	no	
every move	108.6	103.1	−5.55 (−6.70, −4.41)			

Table 2: What separates the bands: points a game given away on each cause at his level (1250–1449) and the band above (1450–1649), the difference Δ (above minus his) with its 95% interval, the Holm-adjusted p, the rule's verdict, and each partition cause's share of the difference on every move. The first seven causes split every move; the next three overlap them.

4.2 Per game and per move

Why would stronger players give away more in quiet positions? A cause's points a game are its points a move times its moves a game, and the band above's games are longer (66.1 plies against 64.4), with more quiet middlegame moves in them (4.1 a game against 3.5). Per move, a check added after the results were read, the band above gives away less on seven of the ten causes and the same, within the interval, on quiet endgames, under a minute on the clock and from move 40 (Table 3); over every move, 3.18 points a move against 3.44. Where the band above gives away more a game, it meets more of those moves; it does not play them worse. The move after the opponent's

serious error costs the most a move in both bands: 7.9 points a move at his level, 2.3 times an average move.

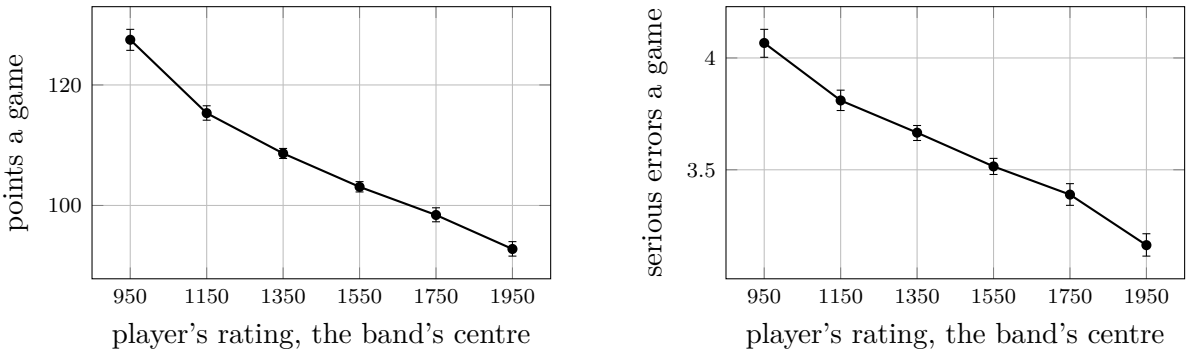
	moves a game		points a move		
cause	his band	above	his band	above	Δ
opening, tense	4.5	4.2	3.07 (3.03, 3.10)	2.66 (2.62, 2.69)	−0.41 (−0.46, −0.36)
opening, quiet	5.8	6.1	2.07 (2.05, 2.09)	1.86 (1.84, 1.88)	−0.21 (−0.24, −0.19)
middlegame, tense	8.5	8.5	4.47 (4.42, 4.51)	4.13 (4.09, 4.17)	−0.33 (−0.39, −0.28)
middlegame, quiet	3.5	4.1	4.17 (4.12, 4.21)	3.87 (3.83, 3.91)	−0.29 (−0.36, −0.24)
endgame, tense	5.6	5.7	3.35 (3.30, 3.39)	3.16 (3.11, 3.21)	−0.19 (−0.26, −0.12)
endgame, quiet	2.9	3.1	3.03 (2.97, 3.09)	2.99 (2.93, 3.05)	−0.04 (−0.12, +0.04)
under a minute left	0.6	0.7	3.71 (3.53, 3.89)	3.49 (3.31, 3.67)	−0.22 (−0.46, +0.02)
tense, under 5 seconds	8.9	9.1	2.67 (2.64, 2.69)	2.45 (2.42, 2.47)	−0.22 (−0.26, −0.18)
after the opponent's error	3.6	3.5	7.88 (7.80, 7.96)	7.35 (7.27, 7.43)	−0.53 (−0.64, −0.42)
from move 40	3.1	3.4	2.90 (2.82, 2.97)	2.88 (2.81, 2.96)	−0.02 (−0.12, +0.09)
every move	31.5	32.4	3.44 (3.42, 3.47)	3.18 (3.16, 3.20)	−0.26 (−0.29, −0.23)

Table 3: Per game and per move (added after the results were read; it tests nothing): each cause's moves a game, and the points a move given away on it, the band's total points over its total moves, with the difference (above minus his).

4.3 Error kinds

Every level gives away fewer points than the one below it, from 127.5 points a game at 850–1049 to 92.8 at 1850–2049, and makes fewer serious errors (Figure 1, Table 4), while its games get longer, from 60 plies to 73. Points a game fall at every band on tense openings and middlegames, quiet openings, fast tense answers and the move after the opponent's serious error, and rise at every band on quiet middlegames and endgames and from move 40 (Table 6).

A player at his level makes 3.67 serious errors a game and one at the band above 3.51. In a check added after the results were read, serious errors carry 94% of the difference between the two bands and hanging a piece 43% on its own (Table 5): his band hangs a piece once every 1.2 games, the band above once every 1.4. Allowing a mate is rare at both levels, mate in one once every 20.8 games at his and once every 23.6 above, and mates carry 13% of the difference. The serious errors the reader leaves untagged, which include every fork or pin that wins material, carry 38%.



(a) Points a game, every move.

(b) Serious errors a game.

Figure 1: Across the six bands, with 95% intervals (most are narrower than the marks).

	per game				hanging a piece:
player's rating	serious errors	hanging a piece	mate in one	a forced mate	once every N games
850–1049	4.07 (4.00, 4.13)	1.02 (1.00, 1.04)	0.072 (0.067, 0.077)	0.072 (0.067, 0.077)	1.0
1050–1249	3.81 (3.76, 3.86)	0.90 (0.89, 0.92)	0.060 (0.057, 0.063)	0.066 (0.062, 0.069)	1.1
1250–1449	3.67 (3.63, 3.70)	0.82 (0.81, 0.83)	0.048 (0.046, 0.050)	0.062 (0.060, 0.065)	1.2
1450–1649	3.51 (3.48, 3.55)	0.74 (0.73, 0.75)	0.042 (0.040, 0.045)	0.055 (0.053, 0.058)	1.4
1650–1849	3.39 (3.34, 3.44)	0.66 (0.65, 0.68)	0.031 (0.028, 0.034)	0.048 (0.044, 0.052)	1.5
1850–2049	3.16 (3.11, 3.21)	0.59 (0.58, 0.61)	0.023 (0.020, 0.026)	0.044 (0.040, 0.048)	1.7

Table 4: Error kinds by band: serious errors a game and the kinds the reader tags among them, with 95% intervals.

	points a game			share of
kind	his band	above	Δ	the difference
serious errors	66.6	61.4	−5.20 (−6.18, −4.23)	94% (87, 101)
hanging a piece	16.6	14.2	−2.38 (−2.70, −2.07)	43% (36, 52)
allowing mate in one	2.4	2.0	−0.38 (−0.54, −0.21)	7% (4, 10)
allowing a forced mate	2.8	2.4	−0.35 (−0.54, −0.18)	6% (3, 10)
other serious errors	44.9	42.8	−2.09 (−2.86, −1.34)	38% (29, 45)

Table 5: What each kind costs, his band against the band above (added after the results were read; it tests nothing): points a game, the difference and its share of the difference on every move. The four kinds below serious errors split them.

4.4 The discount

By the rule, the app's 0.6 is too high on all ten causes: δ runs from 0.10 to 0.33, and is 0.22 over every move (Table 7). The fit cannot carry that verdict. On a calibrated curve a game's result is, near enough, 50 plus the points the opponent gives away less the points the player gives away, and games where one player gives away more are games where the other does too: over every move, the two players' points a game correlate at 0.79. Left out of the fit, the opponent's points pull the slope on the player's points towards zero.

With them in, a check added after the results were read and not part of the rule, a point given away costs 1.05 points of result over every move (1.05 to 1.06), and a point the opponent gives away is worth 1.05 (Table 8). With the seven partition causes in one fit, each beside the opponent's points on it, a point given away costs between 1.02 and 1.25 points of result on every one. The other three causes overlap the partition, so a fit of one of them alone still leaves out points that rise and fall with it, and comes in lower (0.30 for the move after the opponent's serious error).

So the results give no reason to lower the 0.6: a point given away shows in the result one for one, as the curve says it should. What the 0.6 can stand for is the share of a leak a fix removes, and bands of other players cannot measure that.

4.5 His games

His 633 ten-minute games come from the app's own reviews, with its engine and depth, not Lichess's server analysis, so their points are not on the bands' footing. This section compares only how they split between the causes and how often a board shows an error; both were added after the results were read and test nothing.

His points split like his band's on most causes, and two differ (Table 9). Tense endgames take 20.6%

cause	850–1049	1050–1249	1250–1449	1450–1649	1650–1849	1850–2049
opening, tense	24.1	17.6	13.8	11.3	8.5	6.8
opening, quiet	14.1	13.2	12.1	11.3	10.3	9.2
middlegame, tense	49.6	41.9	38.2	35.1	32.6	29.0
middlegame, quiet	12.8	13.6	14.8	15.7	16.7	16.7
endgame, tense	18.6	19.0	18.9	17.9	17.5	16.5
endgame, quiet	6.4	7.7	8.8	9.4	10.0	10.4
under a minute left	2.0	2.2	2.2	2.4	2.9	4.1
tense, under 5 seconds	27.8	25.0	23.8	22.4	20.9	19.0
after the opponent's error	39.7	32.4	28.5	25.5	22.7	19.0
from move 40	5.7	7.4	8.9	9.8	11.4	13.1
every move	127.5	115.3	108.6	103.1	98.4	92.8

Table 6: Points a game on each cause, by band.

cause	δ
opening, tense	0.30 (0.28, 0.32)
opening, quiet	0.29 (0.25, 0.32)
middlegame, tense	0.29 (0.28, 0.30)
middlegame, quiet	0.24 (0.22, 0.26)
endgame, tense	0.22 (0.21, 0.23)
endgame, quiet	0.16 (0.15, 0.18)
under a minute left	0.31 (0.27, 0.35)
tense, under 5 seconds	0.33 (0.32, 0.35)
after the opponent's error	0.10 (0.09, 0.11)
from move 40	0.12 (0.11, 0.13)
every move	0.22 (0.21, 0.23)

Table 7: The rule's discount fit, over his band's player-games: δ, minus the slope of the result on the points given away with the rating difference beside them, and its 95% interval. By the rule every cause is *too high* against the app's 0.6.

of his points, against 17.4% at his level and 17.3% above. Tense openings take 9.6%, against 12.7% at his level, about the band above's 10.9%. He hangs a piece about as often as either band, once every 1.3 games, and allows a mate more often than either: mate in one once every 13.8 games and a forced mate once every 10.0, about 1.6 times his level's rate (Table 10).

cause	correlation of the two players' points	δ with the opponent's points: the cause alone	δ with the opponent's points: every partition cause
opening, tense	0.61	0.78 (0.75, 0.82)	1.06 (1.04, 1.07)
opening, quiet	0.52	0.61 (0.55, 0.67)	1.03 (1.01, 1.06)
middlegame, tense	0.66	0.85 (0.83, 0.87)	1.06 (1.06, 1.07)
middlegame, quiet	0.56	0.53 (0.49, 0.57)	1.07 (1.05, 1.08)
endgame, tense	0.69	0.71 (0.69, 0.74)	1.02 (1.01, 1.03)
endgame, quiet	0.69	0.54 (0.50, 0.58)	1.02 (1.01, 1.04)
under a minute left	0.24	0.41 (0.38, 0.44)	1.25 (1.22, 1.27)
tense, under 5 seconds	0.39	0.55 (0.53, 0.57)	
after the opponent's error	0.67	0.30 (0.28, 0.32)	
from move 40	0.87	0.95 (0.92, 0.98)	
every move	0.79	1.05 (1.05, 1.06)	

Table 8: The discount with the opponent's points (added after the results were read; not part of the rule): the correlation of the two players' points a game, δ with the opponent's points on the same cause beside the player's, and δ in one fit with both players' points on every partition cause.

cause	his games	his band	the band above
opening, tense	9.6% (8.5, 10.8)	12.7% (12.5, 12.9)	10.9% (10.7, 11.1)
opening, quiet	10.3% (9.4, 11.1)	11.1% (11.0, 11.2)	11.0% (10.8, 11.1)
middlegame, tense	36.3% (34.2, 38.6)	35.1% (34.8, 35.5)	34.1% (33.7, 34.4)
middlegame, quiet	13.2% (12.0, 14.3)	13.6% (13.4, 13.8)	15.3% (15.1, 15.5)
endgame, tense	20.6% (18.7, 22.5)	17.4% (17.1, 17.7)	17.3% (17.0, 17.6)
endgame, quiet	7.8% (6.6, 9.1)	8.1% (7.9, 8.3)	9.1% (8.8, 9.3)
under a minute left	2.3% (1.4, 3.3)	2.0% (1.9, 2.1)	2.4% (2.2, 2.5)

Table 9: How the points given away split between the partition's causes (added after the results were read; it tests nothing): his games, his band and the band above, with 95% intervals.

kind	per game: his games	per game: his band	per game: the band above	his games: once every N
hanging a piece	0.777 (0.707, 0.849)	0.823 (0.814, 0.833)	0.738 (0.728, 0.748)	1.3
allowing mate in one	0.073 (0.053, 0.093)	0.048 (0.046, 0.050)	0.042 (0.040, 0.045)	13.8
allowing a forced mate	0.100 (0.073, 0.127)	0.062 (0.060, 0.065)	0.055 (0.053, 0.058)	10.0

Table 10: How often a board shows an error (added after the results were read; it tests nothing): per game in his games and in each band, with 95% intervals.

5 What it means for the app

1. **Where his level loses to the band above.** The ledger's ONE focus picks the cause where he gives away most against his own opponents. The bands add a second yardstick, where players 200 points above give away less, and for his level it puts tense positions first: the middlegame, the opening, then the endgame, with the move after the opponent's serious error and fast answers in tense positions beside them. Quiet middlegames and endgames are not where the band above is better.

2. **A yardstick from other players' games is per move.** Per game, the band above's longer games make quiet positions and late moves look like its weakness. The ledger's own comparison, against the opponent in the same games, meets the same moves and needs no change.

3. **The move after the opponent's error.** It costs 2.3 times an average move at his level and separates the bands, and the think-time study found 33% of serious errors there, on 14% of moves. The coach can teach a check for that move; the think-time study could not show that more time there pays.

4. **Keep the 0.6.** A point given away costs about a point of result. The 0.6 can only discount how much of a leak a fix removes, which needs his own games before and after a focus, not other players' games.

5. **Rates the coach can cite.** At his level a player hangs a piece once every 1.2 games and the band above once every 1.4; his own rate can sit beside them. In his games, mates allowed and tense endgames stand out.

6 Limits

The bands are a cross-section, not a path: the band above is other players, in games against their own level, and their numbers say where stronger players give away less, not what he would gain by fixing one thing. The rule's discount fit adjusted only for the rating difference, and the results show what that left out (Section 4.4); the fits with the opponent's points came after the results were read and are descriptions, not tests. With no engine line, the reader tags only a piece left hanging and a mate; a fork or a pin that wins material goes untagged. Lichess evaluations come from its own server analysis, games someone asked to have analysed are not a random sample, and his Lichess level is the app's conversion of his chess.com rating. His games come from the app's reviews, with another engine and depth, so the paper sets no band's level beside his. Players recur across games and no username is kept, so the intervals, drawn over games, are somewhat narrow.